Should you fold all the three medians to find the centroid? Now you can explore among yourself the following questions:

(i) How can you find the centroid of a triangle?

(ii) Is the centroid equidistant from the vertices?

(iii) Is the centroid of a triangle always in its interior?

(iv) Is there anything special about the medians of an

(a) Isosceles triangle? (b) Equilateral triangle?

Properties of the centroid of a triangle

The location of the centroid of a triangle exhibits some nice properties.

- It is always *located inside the triangle.*
- We have already seen that it *serves as the Centre of gravity* for any triangular lamina.
- Observe the figure given. The lines drawn from each vertex to G form the three triangles ΔABG, ΔBCG, and ΔCAG. Surprisingly, the areas of these triangles are equal.

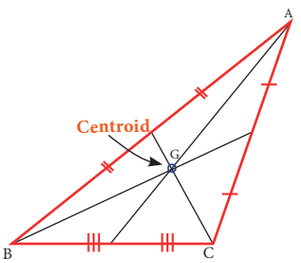
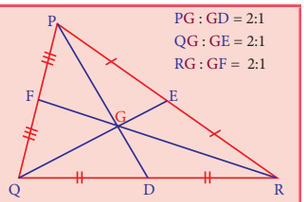


Fig. 5.31

That is, the medians of a triangle divide it into three smaller triangles of equal area!

DO YOU KNOW?

The centroid of a triangle splits each of the medians in two segments, the one closer to the vertex being twice as long as the other one.



This means the centroid divides each median in a ratio of 2:1. (For example, GD is ⅓ of PD). *(Try to verify this by paper folding).*

Example 5.17

In the figure *G* is the centroid of the triangle XYZ.

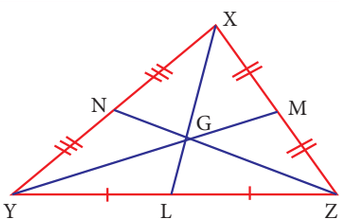


Fig. 5.32

(i) If GL = 2.5 cm, find the length XL.

(ii) If YM = 9.3 cm, find the length GM.

Solution:

(i) Since G is the centroid, XG : GL = 2 : 1 which gives XG : 2.5 = 2 : 1.
Therefore, we get 1 × (XG) = 2 × (2.5) ⇒ XG = 5 cm.
Hence, length XL = XG + GL = 5 + 2.5 = 7.5 cm.

(ii) If YG is of 2 parts then GM will be 1 part. (Why?)
This means YM has 3 parts.
3 parts is 9.3 cm long. So GM (made of 1 part) must be 9.3 ÷ 3 = 3.1 cm.



Example 5.18

ABC is a triangle and G is its centroid. If AD=12 cm, BC=8 cm and BE=9 cm,find the perimeter of ΔBDG.

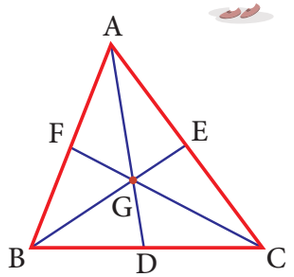


Fig. 5.33

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